BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLYPROGRESSREPORT (08/06/2026 – 14/06/2026)

WEEK 4

PROJECT NAME	RADIO PROPAGATION ANALYSIS AND PATH PREDICTION

PROJECT DESCRIPTION :	Project focuses on simulating antenna beam propagation and predicting communication coverage.
GROUP MEMBER :	GARIMA CHOUDHARY, MANASH PRASAR, ARYAN KUMAR
GROUP ID :	2026G265
GROUP GUIDE :	Dr. ABDUL JHUMMARWALA
GROUP COORDINATOR :	GARIMA CHOUDHARY
COLLEGE GUIDE NAME :	ABHILASHA SAKSENA
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COLLEGE NAME :	GATI SHAKTI VISHWAVIDYALAYA

08/06/2026	• Mapped individual antenna locations as points on the 3D globe within the CesiumJS environment (Garima Choudhary) • Studied the structural properties of standard JSON and GeoJSON formats for geospatial vector data (Manash Prasar) • Structured data arrays representing coordinate pairs for linking transmission nodes (Aryan kumar)
09/06/2026	• Implemented logic in Cesium Sandcastle to dynamically parse coordinates from a custom .json data file. (Garima Choudhary) • Drafted GeoJSON schemas to model geospatial relationships between interconnected antenna stations. (Manash Prasar) • Developed parsing logic to extract geographic coordinates for line-string generation. (Aryan kumar)
10/06/2026	• Successfully connected individual antenna nodes on the map using dynamic polygon lines. (Garima Choudhary) • Validated the parsed GeoJSON vectors to ensure correct spatial rendering on the 3D terrain. (Manash Prasar) • Tested path continuity between mapped data points to ensure proper network line connectivity. (Aryan kumar)
11/06/2026	• Integrated an interactive user console UI element inside the Cesium Sandcastle viewer workspace. (Garima Choudhary) • Added conditional rendering logic to control visibility of specific antenna pattern components. (Manash Prasar) • Programmed console toggles allowing users to selectively filter and isolate the Main Lobe, Side Lobes, or Back Lobes. (Aryan kumar)
12/06/2026	• Explored QGIS workspace configuration and learned how to import vector data layers. (Garima Choudhary) • Studied QGIS interface basics and its application in validating geospatial radio coverage profiles. (Manash Prasar) • Attended a brief overview of QGIS utility for geographic mapping and geospatial file conversions. (Aryan kumar)
13/06/2026	Holiday (2nd Saturday)
14/06/2026	Holiday (Sunday)

Next Week Plan	• **We plan to integrate our processed 3D radiation pattern lobes with the interactive Cesium Sandcastle console UI onto the live map.** • **We will also begin importing our antenna path vector data directly into QGIS to analyze the spatial overlaps against physical terrain datasets.**

REFERENCE:

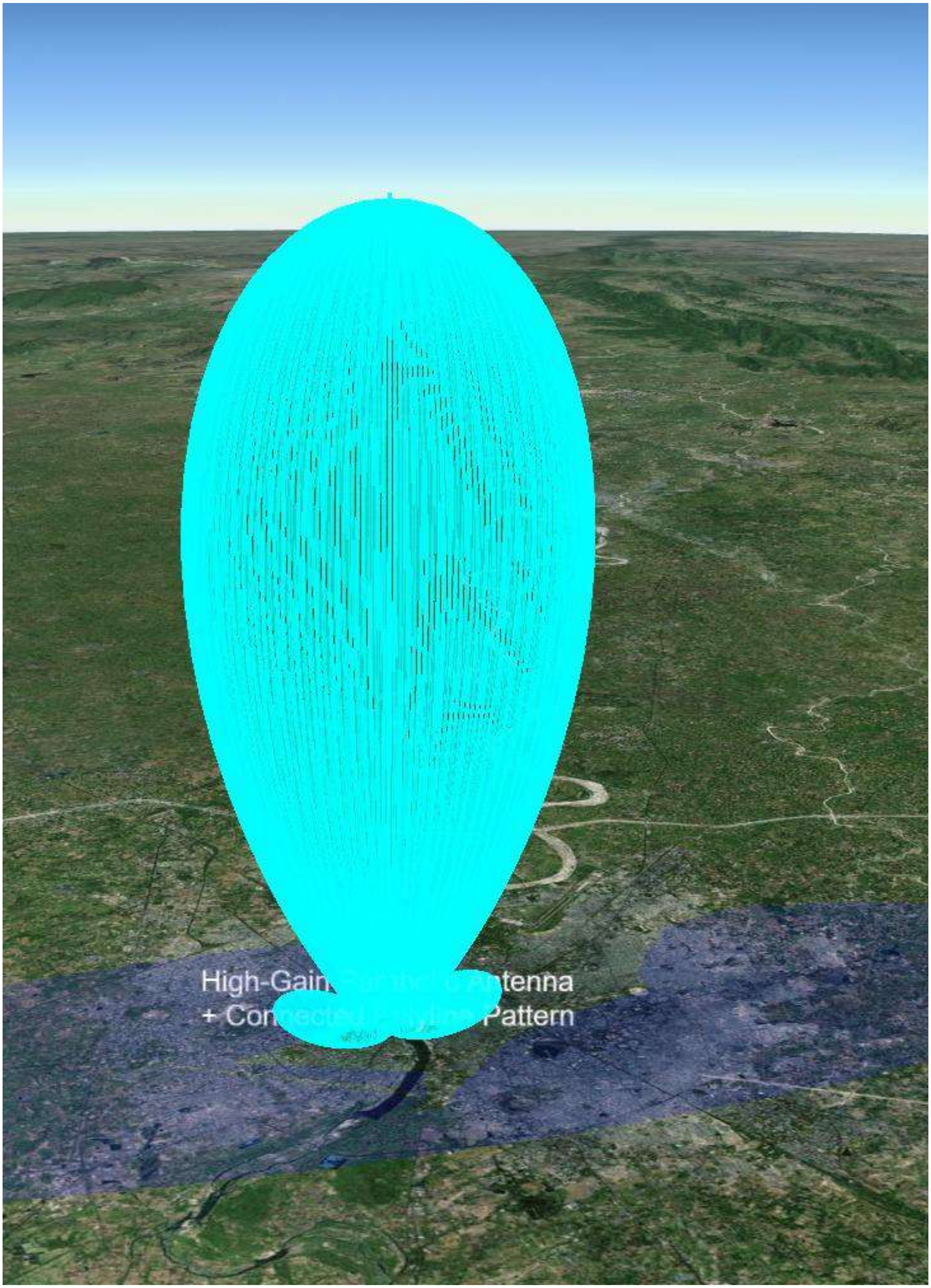
https://sandcastle.cesium.com/

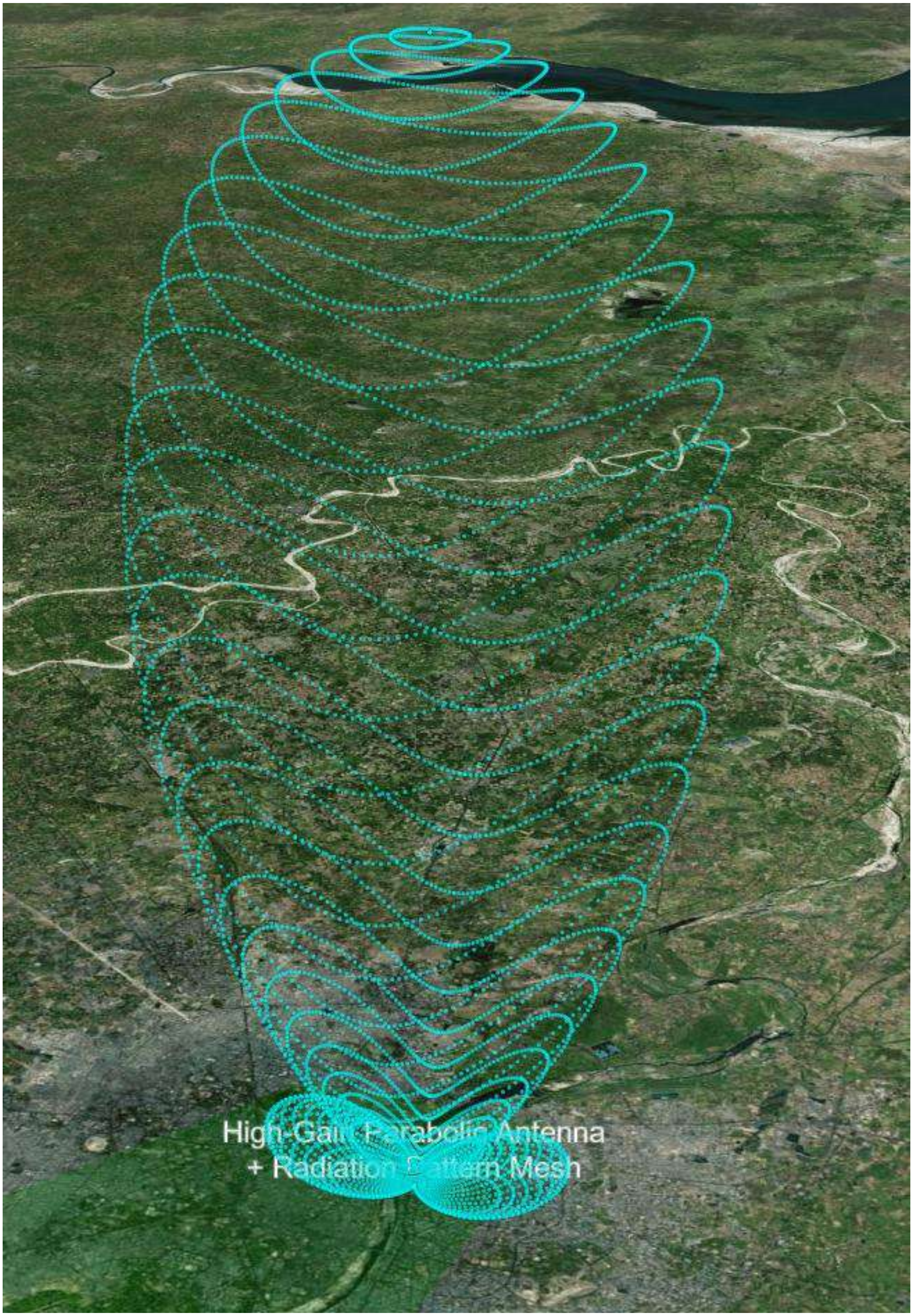
https://cesium.com/learn/cesiumjs-learn/

https://geojson.org/

https://qgis.org/en/site/

SCREEN SHOTS:


High-Gain
Antenna
Pattern


High-Gain Parabolic Antenna
+ Radiation Pattern Mesh